Outliers – observation that the model does not fit well.

- A large residual is not necessarily exceptional, particularly in large datasets.
- we have seen observations can have high influence and this may reduce the size of the residual.
- Consider how large the 'residual' would be if it were not included in the model fit – this leads to **studentised residuals**

Let $\hat{\beta}_{(i)}$ and $\hat{\sigma}^2_{(i)}$ denote the estimates where the i-th observation is excluded. Note that

$$\text{Var}(Y_i - \hat{Y}_{(i)}) = \text{Var}(\underbrace{Y_i - \underline{x}_i\beta}_{\varepsilon_i} + \underline{x}_i\beta - \underline{x}_i\hat{\beta}_{(i)})$$

$$= \text{var}(\varepsilon_i) + \text{var}(\underline{x}_i(\beta - \hat{\beta}_{(i)}))$$

$$= \sigma^2 + \text{Var}(\underline{x}_i\hat{\beta}_{(i)}) = \sigma^2 + \underline{x}_i \,\text{Cov}(\hat{\beta}_{(i)})\,\underline{x}_i^T$$

$$= \sigma^2 + \sigma^2 \underline{x}_i (\underline{X}_{(i)}^T \underline{X}_{(i)})^{-1} \underline{x}_i^T$$

We standardise $y_i - \hat{y}_{(i)}$ replacing σ^2 by $\hat{\sigma}^2_{(i)}$,

$\underline{X}$ is $n \times p$

num of covariates including intercept

$$t_i = \frac{y_i - \hat{y}_{(i)}}{\hat{\sigma}_{(i)}\sqrt{1 + \underline{x}_i(\underline{X}_{(i)}^T\underline{X}_{(i)})^{-1}\underline{x}_i^T}} \sim t_{n-1-p}$$

$$= \frac{\hat{e}_i}{\hat{\sigma}_{(i)}\sqrt{1 - H_{ii}}}$$

$\hat{e}_i$ ← raw residual from full fit

$\hat{\sigma}_{(i)}$ ↖ estimated from $n-1$ observations.

H_{ii} ↖ leverage

We compare the t_i with the quantiles of the t_{n-1-p} distribution.

Suppose there are no outliers – all model assumptions hold. Let A_i be the event that observation i is flagged as an outliers.

$$\mathbb{P}\left(\bigcup_i A_i\right) \leq \sum_i \mathbb{P}(A_i) = n\,\mathbb{P}(A_1)$$

We can set $\mathbb{P}(A_1)$ as α by taking the cut-off to be the $1-\alpha/2$ quantile of the t_{n-p-1} distribution.

We want $n\,\mathbb{P}(A_1) \leq \alpha'$ then $\mathbb{P}(A_1) = \alpha'/n$. This suggests the cut-off $1-\alpha/2n$ quantile of t_{n-p-1}.

Mean Structure

Remember to plot residuals against each covariate, not just fitted values. The plots should show no trend if the model is correct.

Added variable plot. – looks at relationship between response y and covariate x_j after removing the effect of other covariates

Plot of the residuals from $y \sim x_1 + \cdots + x_p$, excluding x_j against the residuals from $x_j \sim x_1 + \cdots + x_p$, excluding x_j.

Partial residual plot

plot of $\hat{e}_i + \hat{\beta}_j x_{ij}$ against x_j.

To handle nonlinearity we could include transformation of the original covariates e.g.

old model $y \sim \beta_1 x_1 + \beta_2 x_2 + \beta_0 + \varepsilon$

new model $y \sim \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \beta_0 + \varepsilon$ (new term)

Collinearity

- If $\underline{\underline{X}}$ does not have full column rank, ie. if there exists a $\underline{c} \in \mathbb{R}^p$ such that $\underline{\underline{X}}\,\underline{c} = \underline{0}$, then $(\underline{\underline{X}}^T\underline{\underline{X}})$ is not invertible and the normal equations

$$(\underline{\underline{X}}^T\underline{\underline{X}})\,\underline{\beta} = \underline{\underline{X}}^T\underline{Y}$$

does not have a unique solution.

- If $\underline{\underline{X}}^T\underline{\underline{X}}$ is close to being singular (ie not invertible) then estimate of β is very imprecise. and the fit is sensitive to measurement errors.

How to detect collinearity?

- Check pairwise correlations. Values close to ± 1 indicate problems, but does not account for complex relationships between covariates.

- Condition number: let $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \cdots \geq \lambda_p > 0$ be the eigenvalues of $\underline{\underline{X}}^T\underline{\underline{X}}$. The condition number

$$\kappa = \sqrt{\frac{\lambda_1}{\lambda_p}}$$

Values ≥ 30 suggest possible problems with collinearity but this is affected by scaling.

- Variance inflation factors — Regress each covariate against all other covariates. Let R_j^2 denote the R^2 valued from regressing j-th covariate on all others

Then

$$\text{var}(\hat{\beta}_j) = \sigma^2 \frac{1}{1 - R_j^2} \frac{1}{\sum_{i=1}^{n} (x_{j,i} - \bar{x}_j)^2}$$

VIF > 5 worth investigating

> 10 suggests problems

Variance inflation factor.

Variance of $\hat{\beta}_j$ if x_j was orthogonal to other covariates

Changes in scale

- Suppose we change a covariate x_j to $(x_j + a)/b$ then the estimated coefficient of β_j changes to $b \cdot \beta_j$ (for non intercept coefficients).
- other statistics such $\hat{\sigma}^2$ and t statistics remain unchanged.

Transforming the response

If you model $\log(y) = \beta_0 + \beta_1 x + \varepsilon$, $\varepsilon \sim N(0, \sigma^2)$

$$\Rightarrow Y = \exp(\beta_0 + \beta_1 x) \cdot \exp(\varepsilon)$$

$$\text{Var}(Y) = \underbrace{\exp(\beta_0 + \beta_1 x)^2}_{= \mathbb{E}(Y)} \cdot \text{Var}(\exp(\varepsilon))$$

$$\mathbb{E}(Y) = \mathbb{E}\left(\exp(\beta_0 + \beta_1 x) \cdot \exp(\varepsilon)\right) = \exp(\beta_0 + \beta_1 x)\, \mathbb{E}(\exp(\varepsilon))$$

$$= \exp(\beta_0 + \beta_1 x) \cdot \exp(\sigma^2/2).$$

So $\text{Var}(Y) = \text{Const} \cdot \mathbb{E}(Y)^2$

Box-Cox transformations

$$g_\lambda(y) = \begin{cases} \dfrac{y^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log(y), & \lambda = 0. \end{cases}$$

Assuming $y > 0$.